



Python Programming - 2301CS404

Lab - 6

Roll No. : 315

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01) WAP to find sum of all the elements in a List.

```
In [22]: li = []  
sum1 = 0  
for i in range(5):  
    li.append(int(input("enter number")))  
    sum1 = sum1 + li[i]  
print(sum1)
```

15

02) WAP to find largest element in a List.

```
In [17]: li = []  
m = 0  
for i in range(5):  
    li.append(int(input("enter number")))  
  
m = max(li)  
print(m)
```

5

03) WAP to find the length of a List.

```
In [18]: li = []  
length = 0
```

```

for i in range(5):
    li.append(int(input("enter number")))
length = len(li)
print(length)

```

5

04) WAP to interchange first and last elements in a list.

```

In [25]: li = []
temp = 0
for i in range(5):
    li.append(int(input("enter number")))
temp = li[0]
li[0] = li[4]
li[4] = temp
print(li)

```

[5, 2, 3, 4, 1]

05) WAP to split the List into two parts and append the first part to the end.

```

In [38]: n = int(input("enter size of list"))
li = []
li1 = []
for i in range(n):
    li.append(float(input("enter number")))
for i in range(n//2, n):
    li1.append(li[i])
for i in range(n//2):
    li1.append(li[i])
print(li1)

```

[4.0, 5.0, 6.0, 1.0, 2.0, 3.0]

06) WAP to interchange the elements on two positions entered by a user.

```

In [29]: first = int(input("enter position : "))
sec = int(input("enter position : "))
li = []
temp = 0
for i in range(6):
    li.append(int(input("enter number")))
temp = li[first]
li[first] = li[sec]
li[sec] = temp
print(li)

```

[2, 1, 3, 4, 5, 6]

07) WAP to reverse the list entered by user.

```
In [39]: li = []
         for i in range(6):
             li.append(int(input("enter number")))
         li.reverse()
         print(li)
```

[6, 5, 4, 3, 2, 1]

08) WAP to print even numbers in a list.

```
In [43]: li = []
         li1 = []
         for i in range(6):
             li.append(int(input("enter number")))
         for i in range(6):
             if(li[i]%2==0):
                 li1.append(li[i])
         print(li1)
```

[2, 4, 6]

09) WAP to count unique items in a list.

```
In [57]: li = []
         count = 0
         for i in range(6):
             li.append(int(input("enter number")))

         for i in range(6):
             if(li.count(li[i])==1):
                 count = count+1
         print(count)
```

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10) WAP to copy a list.

```
In [46]: li = []
         li1 = []
         for i in range(6):
             li.append(int(input("enter number")))
         li1 = li.copy()
         print(li1)
```

[1, 2, 3, 4, 5, 6]

11) WAP to print all odd numbers in a given range.

```
In [47]: li = []
         li1 = []
         for i in range(6):
             li.append(int(input("enter number")))
```

```

for i in range(6):
    if(li[i]%2!=0):
        li1.append(li[i])
print(li1)

```

[1, 3, 5]

12) WAP to count occurrences of an element in a list.

```

In [67]: li = []
count = 0
for i in range(6):
    li.append(int(input("enter number")))
for i in range(6):
    for j in range(i+1,6):
        if(li[i]==li[j]):
            count += 1
print(count)
###print(li[i], "occur" , count , 'times' )

```

2

13) WAP to find second largest number in a list.

```

In [6]: li = []
m = 0
max_sec= 0
for i in range(6):
    li.append(int(input("enter number")))
print(li)
m = max(li)
li.remove(m)
max_sec = max(li)
print('second largest',max_sec)

```

[1, 2, 3, 4, 5, 6]
second largest 5

14) WAP to extract elements with frequency greater than K.

```

In [26]: li = []
li1 = []
f = int(input("enter frequency of your"))
for i in range(5):
    li.append(int(input("enter number")))
for i in li:
    if(f<i):
        li1.append(i)
print(li1)

```

[4, 5]

15) WAP to create a list of squared numbers from 0 to 9 with and without using List Comprehension.

```
In [14]: li = []
li1 = []
ans = []
for i in range(10):
    li.append(int(input("enter number")))
for i in range(10):
    li[i] = li[i]*li[i]
    li1.append(li[i])
print(li1)

### with comprehension
li3 = []
for i in range(10):
    li3.append(int(input("enter number")))
ans = [i**2 for i in li3]
print(ans)
```

```
[0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```

```
[0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```

16) WAP to create a new list (fruit whose name starts with 'b') from the list of fruits given by user.

```
In [18]: li = ['apple' , 'banana' , 'orange']

ans = [i for i in li if i[0] in 'b']
print(ans)
```

```
['banana']
```

17) WAP to create a list of common elements from given two lists.

```
In [22]: li = []
li1 = []
li3 = []
for i in range(3):
    li.append(int(input("enter number")))
for i in range(3):
    li1.append(int(input("enter number")))

for i in li:
    for j in li1:
        if(i==j):
            li3.append(i)
print(li3)
```

```
[2, 3]
```

In []: